COMP4141 Slides Week 3

Brendan Mabbutt

UNSW

March 3, 2026

Admin

Tests Moving Froward

Talk about the test.

- They are 20% in total (one quiz in week $2 - 5$ and $7 - 10$).
- They are Pass level questions
- Only one chance (the 20 minutes during the quiz)
- Based on **the previous week content.**
- Previous week's tutorial questions help with the assessment
- Best 7 of 8

Pumping Lemma & Myhill–Nerode Theorem

Pumping Lemma

Pumping Lemma

If $L \subseteq \Sigma^*$ is regular, then there exists $p \in \mathbb{N}$ (the *pumping length*) such that for every $w \in L$ with $|w| \geq p$, we can write $w = xyz$ satisfying:

1. $|xy| \leq p$,
2. $|y| > 0$,
3. $xy^iz \in L$ for all $i \in \mathbb{N}$.

$w^i = \underbrace{ww\ldots w}_{i \text{ time}}$

$xz \in L$ (Pump down)

$xy^2z \in L$ (Pump up)

Myhill–Nerode Theorem

Distinguishability

Let $L \subseteq \Sigma^*$. $x, y \in \Sigma^*$

We say x and y are *distinguishable by* L if there exists $z \in \Sigma^*$ such that:

$$xz \in L \quad \text{and} \quad yz \notin L$$

(or vice versa).

We write $x \equiv_L y$ if x and y are not distinguishable by L.

The *index* of L is the number of $\equiv_L$-equivalence classes.

Myhill–Nerode Theorem

A language $L \subseteq \Sigma^*$ is regular if and only if the index of L is finite.

Moreover, this index equals the minimum number of states of any DFA that accepts L.

Using Myhill–Nerode

Myhill–Nerode proof format

Find an infinite sequence of words (not necessarily in L), $u_1, u_2, u_3, \ldots,$ and a doubly indexed sequence of words $w_{i,j}$ for $i < \cancel{j} \in \mathbb{N}_0$ such that

$$u_i w_{i,\cancel{j}} \in L \quad \text{and} \quad u_j w_{i,\cancel{i}} \notin L$$

(or vice versa).

Context Free Grammars

Formal Definition

Definition

A *context-free grammar* (*CFG*) is a 4-tuple (N, Σ, P, S), where

1. N is a finite set of *variables*,
2. Σ is a finite set, disjoint from N, of *terminals*,
3. $P \subseteq N \times (N \cup \Sigma)^*$ is a finite set of *rules*, and
4. $S \in N$ is the *start variable*.

Variables are often called *non-terminal symbols*, terminals are often called *terminal symbols*, rules also go under the name *productions*, and the start variable is also known as the *sentence symbol*.

Derivations

The language of a given CFG, $G = (N, \Sigma, P, S)$, can be characterized using the concept of a *derivation*.

Definition

Derivation step: $\alpha A \beta \Rightarrow_G \alpha \gamma \beta$ whenever $A \to \gamma \in P$.

Define $\Rightarrow_G^*$ to be the *reflexive transitive closure* of $\Rightarrow_G$. That is, $\alpha \Rightarrow_G^* \beta$ if we can get from α to β in zero or more steps.

The *language of G* is

$$L(G) = \left\{ w \in \Sigma^* \;\middle|\; S \Rightarrow_G^* w \right\}$$

Parse Tree

Definition

A **parse tree** is a tree that shows how to derive a string from a non-terminal.

Definition

The concatenation of the symbols at the leaves of a parse tree is called the **yield** of the parse tree.

Leftmost Derivations

Definition

A derivation of a string w in a grammar G is a **leftmost derivation** if at every step the leftmost remaining variable is the one replaced.

Definition

A context-free grammar (CFG) is **ambiguous** if there is more than one leftmost derivation for the same string.

Pushdown Automata

Definition

A **pushdown automaton (PDA)** is a 6-tuple $(Q, \Sigma, \Gamma, \delta, q_0, F)$ where:

1. Q is the set of states,
2. Σ is the input alphabet,
3. Γ is the stack alphabet,
4. $\delta : Q \times (\Sigma \cup \{\epsilon\}) \times (\Gamma \cup \{\epsilon\}) \to 2^{Q \times \Gamma^*}$ is the transition relation,
5. $q_0 \in Q$ is the start state, and
6. $F \subseteq Q$ is the set of accept states.

Acceptance by a PDA

Definition

An **instantaneous description (ID)** is a snapshot (q, w, α), where:

- $q \in Q$ is the current state,
- $w \in \Sigma^*$ is the portion of the input not yet read,
- $\alpha \in \Gamma^*$ is the complete contents of the stack.

A string w is **accepted** if there exists $\gamma \in \Gamma^*$ and $p \in Q$ such that

$$(q_0, w, \epsilon) \rightsquigarrow^* (p, \epsilon, \gamma), \quad p \in F.$$

Power of Pushdown Automata

Theorem

A language $L \subseteq \Sigma^*$ is **context-free** if and only if some PDA recognizes L.

W3 P1

Problem 1

Show that the following languages are not regular:

(a) $\{0^i 1^j : 0 \leq i \leq j\}$

$\parallel$

L

Suppose L is regular. Then by PL we have a $p \in \mathbb{N}$

Assume ① ② hold $\forall w \in L : |w| \geq p$

then we need ③

Try:

$0 0 1^{p-2}$

(can't do $0^{p-2} 1^2$ if $p \geq 3$)

Exc: you can this when $p \geq 5$ (my guess)

Try: $0^p 1^p$

$$\overbrace{00\ldots0}^{m\ 0\text{'s}}\ldots0\underbrace{1\ldots1}_{p\ 1\text{'s}}$$

p 0s

If $xcyz = 0^p 1^p$

by 1 of PL we have

$|xcy| \leq p$

so $xy = 0^m$ $m \leq p$

So by ② of PL we need

$|y| > 0$ so $y = 0^k$ for $k > 0$

$x = 0^{m-k}$

$y = 0^k$

$z = 0^{p-m} 1^p$

Try to pump up

$xyz = 0^{m-k} 0^{2k} 0^{p-m} 1^p$

$= 0^{p+k} 1^p$

$\notin$ Contradiction

Pump down

$xz = 0^{m-k} 0^{p-m} 1^p$

$= 0^{m-k} 1^p$

Problem 1

Show that the following languages are not regular:

(a) $\{0^i 1^j : 0 \leq i \leq j\}$

We want $u_1, u_2, \ldots$

$u_j = 0^j$

Want that if $i < k$

want a $w_{i,k}$ s.t.

$\sigma^i w_{i,k} \notin L$ and $\sigma^k w_{i,k} \in L$

(or vice versa)

Have $w_{i,k} = 1^i$

then we $0^i 1^i \in L$ and $0^k 1^i \notin L$

$(k > i)$

(would not work if one did $w_{i,k} = 1^k$)

Problem 1

Show that the following languages are not regular:

(b) $\{w \in \{0,1\}^* : \text{the number of 0's in } w \text{ is equal to the number of 1's in } w\}$

Problem 1

Show that the following languages are not regular:

(c) $\{ww : w \in \{0, 1\}^*\}$

W3 P2

Problem 2

Give CFGs that generate the following languages:

(a) $\{0^i 1^j : 0 \leq i \leq j\}$

$S \to 0S1 \mid T$

$T \to T1 \mid 1$

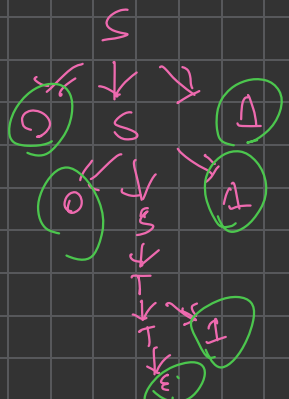
$00111 \in L$

$S \Rightarrow_G 0S1 \Rightarrow_G 00S11 \Rightarrow_G 00T11$

$\Rightarrow_G 00T111$

$\Rightarrow_G 00111$

Parse Tree



00111

Problem 2

Give CFGs that generate the following languages:

(b) $\{w : w \text{ contains } 00\}$

$S \to T00T$

$T \to 0T \mid 1T \mid \varepsilon$

Problem 2

Give CFGs that generate the following languages:

(c) $\{0^i 1^j 0^j 1^i : i, j \geq 0\}$

$S \to 0S1 \mid T$

$T \to 1T0 \mid \varepsilon$

W3 P3

Problem 3 (Part 1)

Consider the following context free grammar for a simple programming language: $G = (N, \Sigma, P, S)$, where $N = \{S, I, A, C, E, V\}$, $\Sigma = \{x,\ y,\ z,\ <,\ :=,\ ;,\ \{,\ \},\ \textbf{if},\ \textbf{then},\ \textbf{else}\}$. Here the inner braces are terminal symbols and the outer braces are part of the set notation. Treat the keywords **if**, **then**, **else** as if each is a single terminal symbol rather than a sequence of letters.

Problem 3 (Part 2)

(a) Show the parse tree for a simple program in $L(G)$ that sorts the variables x, y into ascending order.

Here P contains the following productions:

$$S \to I \mid A \mid C$$
$$I \to \textbf{if } E \textbf{ then } S$$
$$I \to \textbf{if } E \textbf{ then } S \textbf{ else } S$$
$$A \to V := V$$
$$C \to \{S; S\}$$
$$E \to V < V$$
$$V \to x \mid y \mid z$$

Want to produce

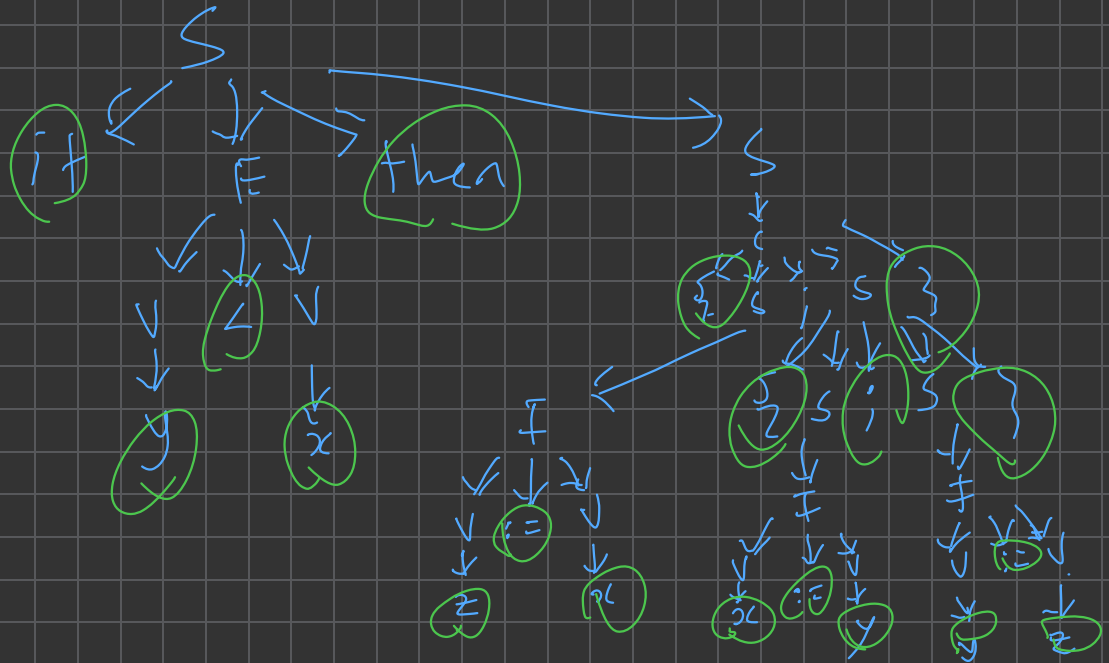
if $y < x$ then $\{z := x\}$ $\{x := y;\ y :=$

$S \rightarrow$ if E then S

$\rightarrow$ if $V < V$ then C

$\rightarrow$ if $y < x$ then $\{S; S\}$

$\vdots$


S
if
E
then
S
y
<
x
{
S
}
S
;
S
z
:=
x
:=
y
}
z

Problem 3 (Part 2)

(b) Show that this grammar is ambiguous, by showing the parse trees for a string in the language that has two distinct parse trees. (Hint: consider nested uses of the I productions.)

Here P contains the following productions:

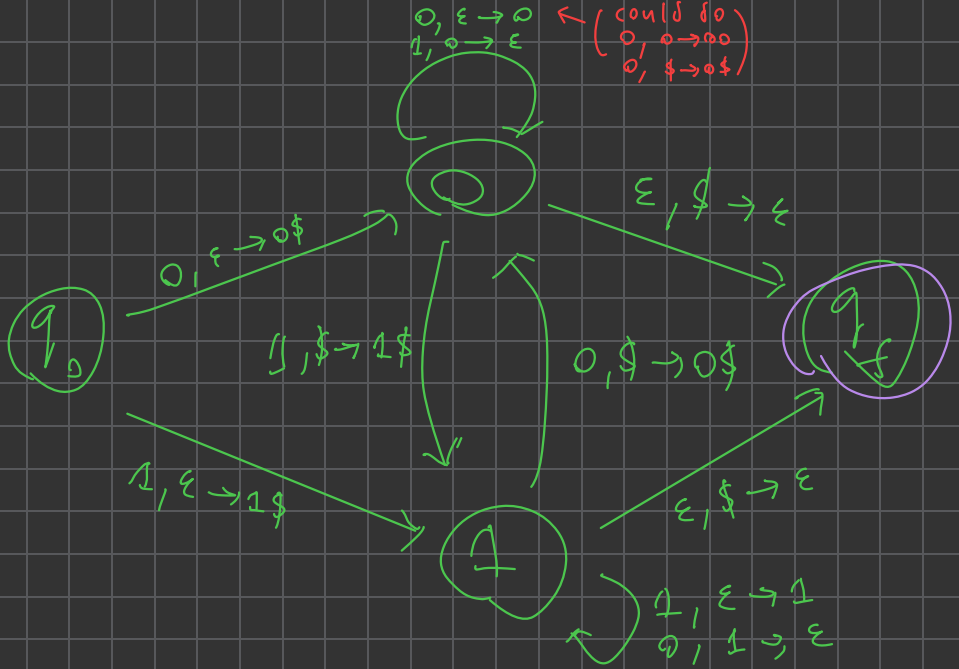
$$
\begin{aligned}
S &\to I \mid A \mid C \\
I &\to \textbf{if } E \textbf{ then } S \\
I &\to \textbf{if } E \textbf{ then } S \textbf{ else } S \\
A &\to V := V \\
C &\to \{S; S\} \\
E &\to V < V \\
V &\to x \mid y \mid z
\end{aligned}
$$

W3 P4

Problem 4

Give PDAs that recognise the following languages:

(a) $\{w \in \{0,1\}^* : \text{the number of 0's in } w \text{ is equal to the number of 1's in } w\}$


0, ε → 0
1, 0 → ε
(could do
0, 0 → 00
0, $ → 0$)
0, ε → 0$
1, $ → 1$
0, $ → 0$
ε, $ → ε
1, ε → 1$
ε, $ → ε
1, ε → 1
0, 1 → ε
q0
0
1
qf

Problem 4

Give PDAs that recognise the following languages:

(b) $\{0^i 1^j 0^j 1^i : i, j \geq 0\}$

Problem 4

Give PDAs that recognise the following languages:

(c) $\{w \in \{0,1\}^* : \text{the number of 0's in } w \text{ is twice the number of 1's in } w\}$